Sunbeam Institute of Information Technology
Pune and Karad

Module - Concepts of Operating Systems and Administration

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Types of Kernel

1. Monolithic Kernel (BSD UNIX)
2. Micro Kernel (Symbian, MACH)
3. Modular Kernel (Windows)
4. Nano Kernel (FreeRTOS)
5. Hybrid Kernel (MacOS - Darwin : BSD UNIX + MACH)

Linux Distribution

Operating system

- Linux Kernel
- Boot loader (LiLo / GrUB) (UBoot - embedded linux)
- File system (ext4)
- Package manager (apt / apt-get / snap)
- . . . . .

Linux Kernel:

Monolithic component

1. Process Management
2. CPU scheduling
3. Memory management
4. Hardware abstraction
5. System calls
6. IO Management

...

(vmlinuz)

Modular component

1. File system mgrs
2. Device drivers

....

dynamically loadable modules

.ko → kernel object

Micro component

GUI

xserver ← Process server

User Interfacing

CLI (Command Line Interface)

- is provided with the help of shell.
- shell is a program which acts as a intermediate between OS & user
- shell is also called as command interpretor.

Windows : cmd.exe , powershell.exe
Linux : bsh (sh), bash, csh, ksh, zsh

bash → Bourne Again Shell (Default shell)

GUI (Graphical User Interface)

- provided as seperate process servers
- xServer

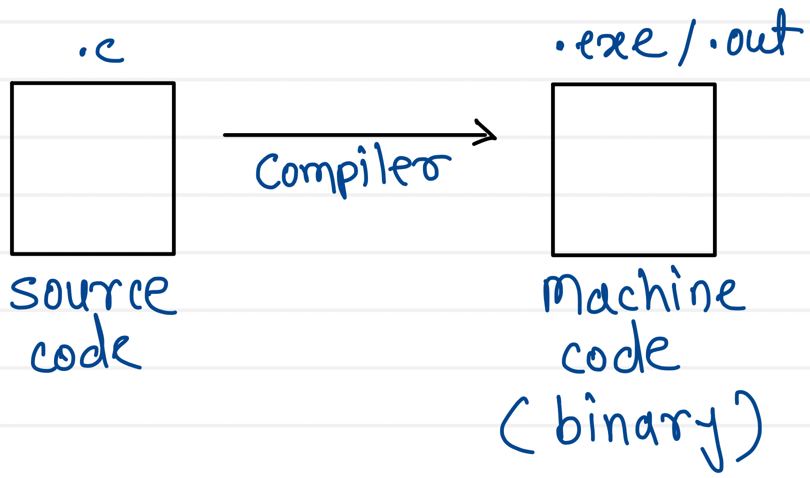
Windows : explorer.exe
Linux :
1. GNOME (GTK) (GNU Network Object Model Enviroment)
2. KDE plasma (Qt) (K Desktop Enviroment)
3. XFCE (XForm) (XForms Common Enviroment)

tool kits (libraries) to create GUI

Program compilation steps

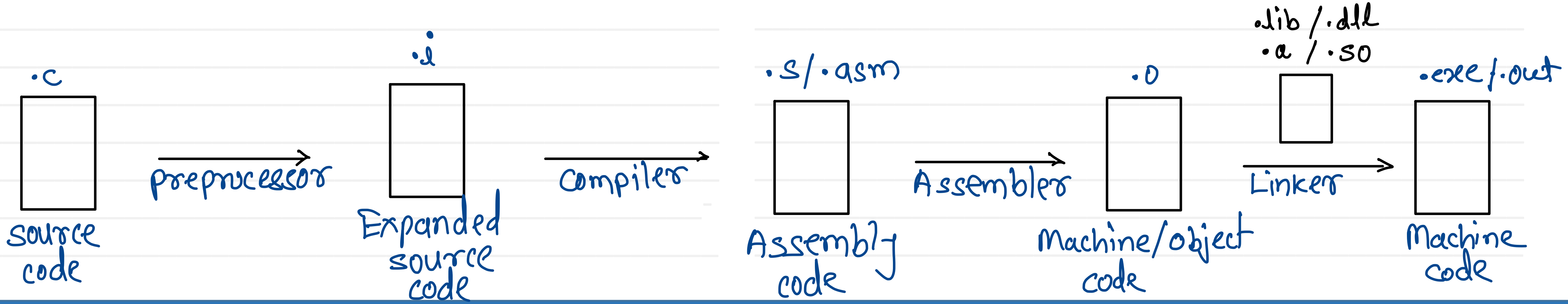
GCC – GNU C Compiler
GNU – GNU Not UNIX

Process : Program in execution
Program : set of instructions for machine (CPU)



Toolchain : (gcc)
– set of tools which work on source code one by one to convert it into machine code.

1. Preprocessor (cpp)
2. Compiler (cc)
3. Assembler (as)
4. Linker (ld)
5. Debugger (gdb)
6. Utilities (make, objdump)



Program

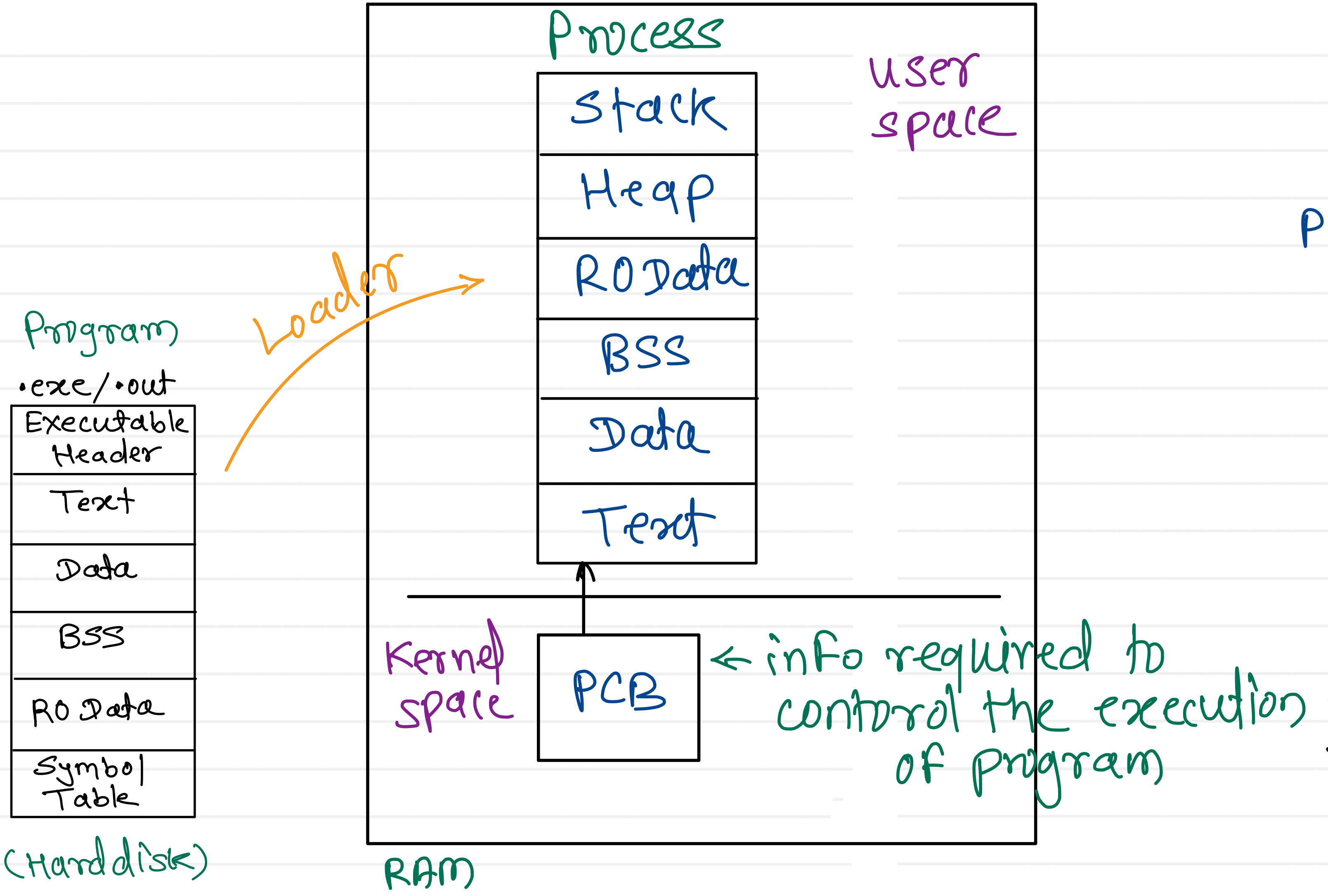
.exe/.out

Executable Header	- Magic number (2 or 4 bytes) - identity to file format. Windows - .exe - Portable Executabe (PE) — MZ Linux — .out - Executable Linking Format (ELF) — 2ELF - type of program (CLI/GUI/library) - info about remaining sections (start, end, size) - address of entry point function.
Text	- instructions of program in binary format
Data	- static & global variables (initialized)
BSS	- static & global variables (uninitialized)
RO Data	- read only data (string constant)
Symbol Table	- info about symbols of a program

(sectioned binary)

Symbols
- variables - type, size, address, section.....
- functions - return type, address, no.of/type of args

Process

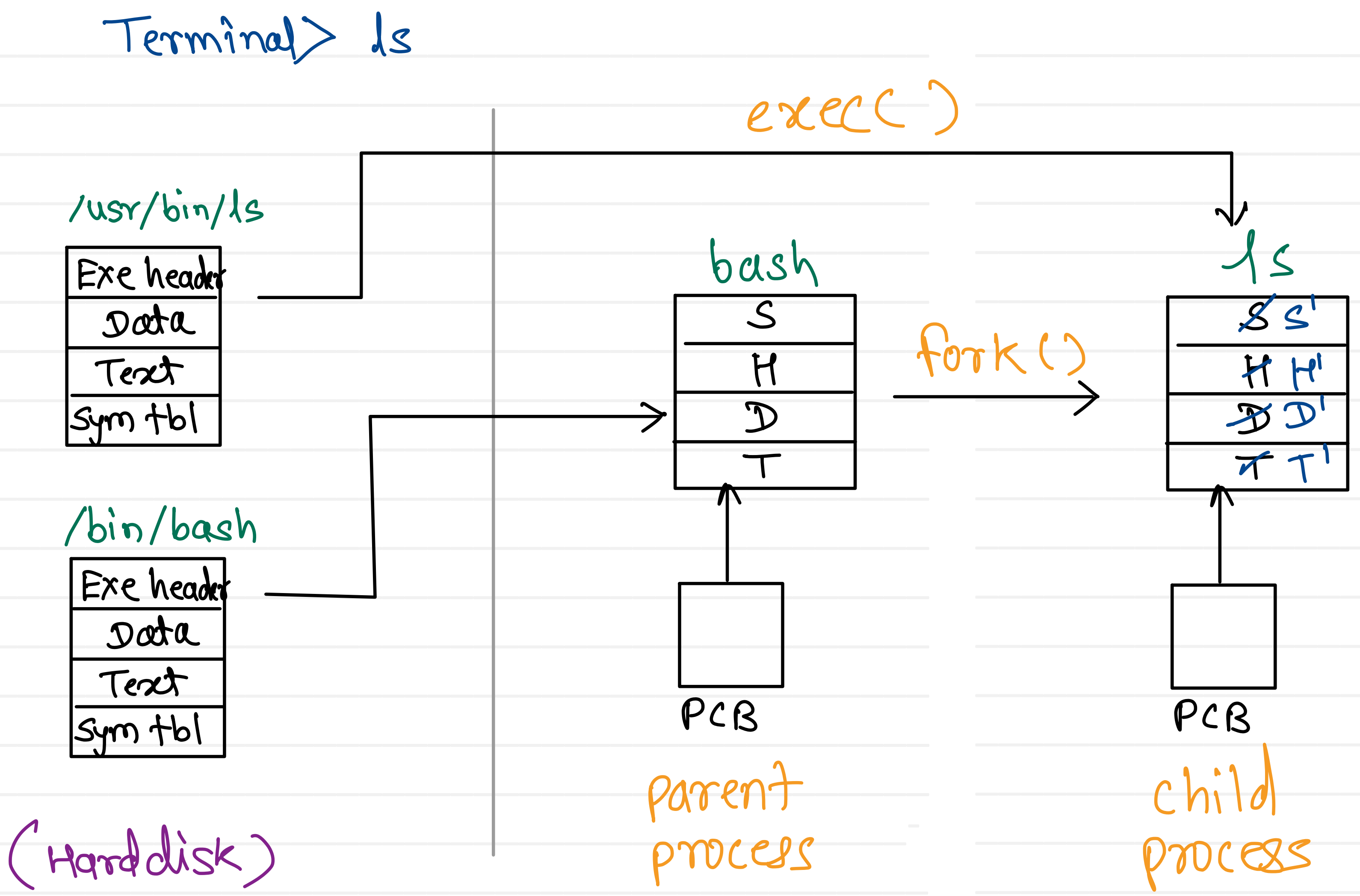


To read executables :
readelf / objdump
xxd

PCB – process descriptor
↓
struct task_struct
(sched.h)

- pid , ppid
- exit status
- CPU sched info
- Mem info
- IPC info
  - message queue
  - pipe
  - signals
  - socket
  - shared memory
- File info
....

Shell internals - how commands executes?



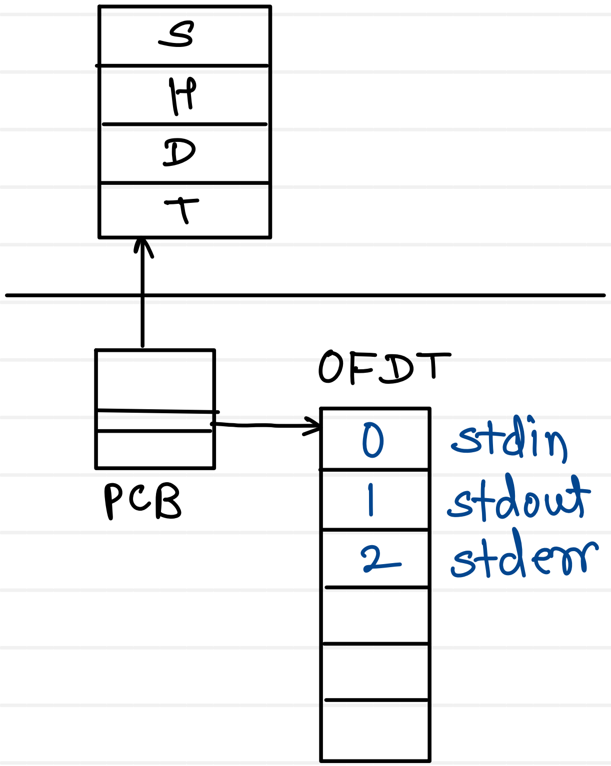
Redirection

- by default, every program
  takes input from standard input (stdin)
  writes output on standard output (stdout)
  writes error on standard error (stderr)

- we can change this default behaviour
  by doing 'redirection'

- There are three types of redirections
  1. Output redirection (>) (>>)
     instead of writing output on stdout it is written into file
  2. input redirection (<)
     instead of taking input from stdin, it is taken from file
  3. error redirection (2>) (2>>)
     instead of writing error on stderr it is written into file

e.g. cat



Pipe

- pipe can be used to do nesting of multiple commands
- output of first command is given to input of second command.
- pipe is a unidirectional IPC mechnism